

Lab Selection I

1- Given the following definitions:

`x, y, z = 3, 5, 7` #this is a correct assignment in Python

evaluate the following Boolean expressions:

- a. `x == 3` True
- b. `x < y` True
- c. `x >= y` False
- d. `x <= y` True
- e. `x != y - 2` False
- f. `x < 10` True
- g. `x >= 0`
- h. `x < 0 and x < 10`
- i. `x >= 0 and x < 2`
- j. `x < 0 or x < 10`
- k. `x > 0 or x < 10`
- l. `x < 0 or x > 10`

2- Given the following definitions:

`x, y = 3, 5`

`b1, b2, b3, b4 = True, False, x == 3, y < 3`

evaluate the following Boolean expressions:

- a. `b3`
- b. `b4`
- c. `not b1`
- d. `not b2`
- e. `not b3`
- f. `not b4`
- g. `b1 and b2`
- h. `b1 or b2`
- i. `b1 and b3`
- j. `b1 or b3`
- k. `b1 and b4`
- l. `b1 or b4`
- m. `b2 and b3`
- n. `b2 or b3`
- o. `b1 and b2`
- p. `b1 or b2 and b3`
- q. `b1 and b2 and b3`
- r. `b1 or b2 or b3`
- s. `not b1 and b2 and b3`

- t. not b1 or b2 or b3
- u. not (b1 and b2 and b3)
- v. not (b1 or b2 or b3)
- w. not b1 and not b2 and not b3
- x. not b1 or not b2 or not b3
- y. not (not b1 and not b2 and not b3)
- z. not (not b1 or not b2 or not b3)

3- Express the following Boolean expressions in simpler form; use fewer operators.
 x is an integer.

- a. not (x == 2)
- b. x < 2 or x == 2
- c. not (x < y)
- d. not (x <= y)
- e. x < 10 and x > 20
- f. x > 10 or x < 20
- g. x != 0
- h. x == 0

4- Express the following Boolean expressions in an equivalent form without the not operator.
 x and y are integers.

- a. not (x == y)
- b. not (x > y)
- c. not (x < y)
- d. not (x >= y)
- e. not (x <= y)
- f. not (x != y)
- g. not (x == y and x < 2)
- h. not (x == y or x < 2)
- i. not (not (x == y))